

Faculty of Information Technology
Delta Technology University (DTU)

MediScan AI

An Intelligent Medical Diagnosis Platform

Graduation Project Report

Submitted in Partial Fulfillment of the Requirements
for the Bachelor's Degree in Computer Science

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Dedication

To our families, whose love and sacrifices made this journey possible.

To our professors, whose guidance shaped our thinking.

And to every patient who deserves faster, more accurate diagnosis.

Acknowledgment

We would like to express our sincere gratitude to the Faculty of Information Technology at Delta Technology University (DTU) for providing us with the academic foundation and resources necessary to complete this project.

We are deeply thankful to our project supervisor for their continuous support, insightful feedback, and invaluable guidance throughout the research and development phases of this work.

Special thanks go to the open-source community and the creators of the datasets used in this project, including Kaggle contributors who made publicly available medical imaging data that was crucial for training and validating our models.

Finally, we acknowledge the collective effort of our entire team. The success of this project is a testament to our collaboration, perseverance, and shared commitment to applying technology for a meaningful purpose.

Abstract

Artificial intelligence and deep learning have emerged as transformative tools in modern healthcare, enabling computer-aided diagnosis systems that rival expert-level accuracy. This project presents MediScan AI, a unified intelligent medical diagnosis platform designed to assist clinicians in the early detection of six critical medical conditions using both medical imaging and clinical laboratory data.

The platform integrates two categories of diagnostic models. The first category employs Deep Learning (DL) models based on the ResNet-18 convolutional neural network architecture, fine-tuned on curated medical imaging datasets for the detection of Brain Tumors (MRI), Brain Stroke (CT scan), Diabetic Retinopathy (fundus photographs), and Chest X-Ray abnormalities including Pneumonia. The second category utilizes classical Machine Learning (ML) pipelines for the prediction of Chronic Kidney Disease (CKD) stages and Diabetes, based on structured clinical biomarker data.

A key feature of the platform is the integration of Gradient-weighted Class Activation Mapping (Grad-CAM), which generates visual saliency maps overlaid on medical images. These heatmaps highlight the anatomical regions that most influenced the model's prediction, providing physicians with interpretable and explainable AI output — a critical requirement for clinical trust and regulatory acceptance.

The backend infrastructure is built with FastAPI, offering a production-grade RESTful API server that handles image uploads, model inference, and structured prediction responses. A responsive web frontend provides an intuitive interface for both patients and healthcare providers, featuring dedicated dashboards, diagnostic modules, and diagnostic report generation.

Experimental results demonstrate high classification accuracy across all modules, with the brain tumor classifier achieving over 94% accuracy and the diabetes prediction model exceeding 97% accuracy on test sets. These results validate the clinical viability of the proposed system as a decision-support tool in medical practice.

Keywords: *Medical AI, Deep Learning, ResNet-18, Grad-CAM, CNN, Brain Tumor, Brain Stroke, Diabetic Retinopathy, Chest X-Ray, CKD, Diabetes, FastAPI, Computer-Aided Diagnosis.*

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Chapter 1: Introduction

1.1 Background and Motivation

The intersection of artificial intelligence and healthcare has created unprecedented opportunities for improving diagnostic accuracy, reducing clinical workload, and democratizing access to expert-level medical analysis. Globally, the shortage of radiologists and specialist physicians — particularly in developing regions — creates diagnostic bottlenecks that lead to delayed treatment and preventable patient harm.

Deep Learning (DL), a subfield of machine learning inspired by the human brain's neural architecture, has demonstrated remarkable capability in image recognition tasks. In the medical domain, convolutional neural networks (CNNs) have been trained on thousands of annotated medical images to detect pathologies that are often subtle and easily missed by the untrained eye.

MediScan AI was conceived with a clear mission: to build a production-ready, unified diagnostic platform capable of analyzing both medical images and clinical data, presenting results in a clinically interpretable format that aids — rather than replaces — the physician's judgment.

1.2 Problem Statement

Despite technological advances, several critical challenges persist in AI-driven medical diagnosis:

- **Fragmented solutions:** existing AI diagnostic tools typically address only a single disease or modality, requiring clinicians to use multiple disconnected systems.
- **Lack of interpretability:** most deep learning models operate as 'black boxes,' producing predictions without explaining which image features drove the decision — creating a barrier to clinical adoption.
- **Limited accessibility:** state-of-the-art diagnostic AI is often locked behind expensive proprietary platforms, inaccessible to smaller healthcare facilities.
- **Integration complexity:** deploying AI models in a clinical workflow requires robust, scalable backend infrastructure that can handle concurrent requests and diverse data formats.

MediScan AI directly addresses all four of these challenges through a unified, explainable, open-architecture platform.

1.3 Project Objectives

The primary objectives of this project are:

1. Design and implement a multi-disease AI diagnostic platform supporting six medical conditions.
2. Train deep learning models (ResNet-18) for four imaging-based diagnostic tasks with high accuracy.
3. Develop machine learning pipelines for two clinical data-based prediction tasks.
4. Integrate Grad-CAM visualization to provide transparent, explainable AI predictions.
5. Build a FastAPI-based RESTful backend to serve all models in production.
6. Develop an intuitive web-based frontend for both patients and healthcare providers.

1.4 Project Scope

The scope of MediScan AI encompasses the following six diagnostic modules:

Module	Type	Input	Output
Brain Tumor	Deep Learning	MRI Image	4-class tumor type
Brain Stroke	Deep Learning	CT Scan Image	3-class stroke type
Diabetic Retinopathy	Deep Learning	Fundus Photo	Binary: DR / No DR
Chest X-Ray	Deep Learning	X-Ray Image	Binary: Pneumonia / Normal
CKD Staging	Machine Learning	Clinical Labs	CKD Stage (1–5)
Diabetes	Machine Learning	Clinical Labs	Diabetic / Non-Diabetic

1.5 Report Organization

The remainder of this report is structured as follows:

- Chapter 2 reviews related literature and prior work in medical AI.
- Chapter 3 describes the overall system architecture and design decisions.
- Chapter 4 presents the deep learning modules in detail.
- Chapter 5 covers the machine learning modules.
- Chapter 6 describes the datasets used for training and evaluation.
- Chapter 7 presents experimental results and performance analysis.
- Chapter 8 showcases the system interface and user experience.
- Chapter 9 concludes the report and discusses future directions.

Chapter 2: Literature Review

2.1 Medical Imaging AI: An Overview

Artificial intelligence in medical imaging has evolved significantly over the past decade. Early computer-aided diagnosis (CAD) systems relied on hand-crafted features and traditional classifiers such as support vector machines (SVM) and random forests. These systems, while groundbreaking for their time, required extensive domain expertise to engineer meaningful features from raw pixel data.

The advent of deep learning — particularly convolutional neural networks (CNNs) — fundamentally changed this paradigm. CNNs learn hierarchical feature representations directly from raw images, eliminating the need for manual feature engineering. Landmark works such as AlexNet (Krizhevsky et al., 2012), VGGNet (Simonyan & Zisserman, 2014), and ResNet (He et al., 2016) demonstrated that sufficiently deep networks with residual connections could achieve superhuman performance on image classification benchmarks.

2.2 Convolutional Neural Networks in Medicine

CNNs have been successfully applied across virtually every modality of medical imaging:

- Radiology: CNNs have matched or exceeded radiologist performance in detecting pulmonary nodules on CT scans and diagnosing pneumonia from chest X-rays (Rajpurkar et al., 2017 — CheXNet achieved radiologist-level pneumonia detection).
- Ophthalmology: Google's DeepMind demonstrated CNN-based detection of diabetic retinopathy and macular degeneration from retinal photographs with accuracy surpassing specialist ophthalmologists.
- Neurology: CNN models trained on brain MRI have demonstrated capability in classifying gliomas, meningiomas, and pituitary tumors.
- Pathology: Digital pathology systems use CNNs to analyze histological slides for cancer grading.

2.3 Transfer Learning Approaches

A central challenge in medical AI is data scarcity. Annotated medical imaging datasets are expensive to produce, requiring expert radiologists to label thousands of images. Transfer learning addresses this by leveraging models pre-trained on large general-purpose datasets (typically ImageNet with over 1.2 million images) and fine-tuning them on smaller domain-specific datasets.

ResNet-18 (Residual Network with 18 layers), developed by He et al. at Microsoft Research, is particularly well-suited for medical imaging transfer learning due to its:

- Residual skip connections that mitigate the vanishing gradient problem in deep networks.

- Compact architecture (11.7 million parameters) balancing accuracy and computational efficiency.
- Strong ImageNet pre-trained weights that capture general visual patterns (edges, textures, shapes) transferable to medical images.

2.4 Explainable AI (XAI) in Healthcare

Despite impressive accuracy, the 'black box' nature of deep learning models poses a fundamental challenge for clinical deployment. Clinicians require not just a prediction, but an understanding of why the model made that prediction — both for trust and for regulatory compliance.

Gradient-weighted Class Activation Mapping (Grad-CAM), introduced by Selvaraju et al. (2017), addresses this by using the gradients of a target class flowing into the final convolutional layer to produce a coarse localization map highlighting the important regions in the image that influenced the model's decision. In a medical context, a Grad-CAM heatmap overlaid on a brain MRI allows a radiologist to verify that the model is attending to the correct anatomical region — not an imaging artifact or background noise.

2.5 Related Work

The following table summarizes key prior work in AI-based medical diagnosis:

Author(s)	Year	Task	Model	Accuracy
Rajpurkar et al.	2017	Chest X-Ray Pneumonia	CheXNet (DenseNet-121)	91.0%
Gu et al.	2019	Brain Tumor MRI	CE-Net (CNN)	88.4%
Gulshan et al.	2016	Diabetic Retinopathy	Inception-v3	97.5% AUC
Chen et al.	2020	Brain Stroke CT	ResNet-50	92.1%
Qin et al.	2021	CKD Prediction	Random Forest	95.0%
This Project	2025	6 Diseases (unified)	ResNet-18 + ML	94–97%

The primary differentiator of MediScan AI from all prior works is its unified multi-disease architecture, production-ready API infrastructure, and integrated Grad-CAM explainability — making it not just a research prototype but a deployable clinical decision-support system.

Chapter 3: System Architecture

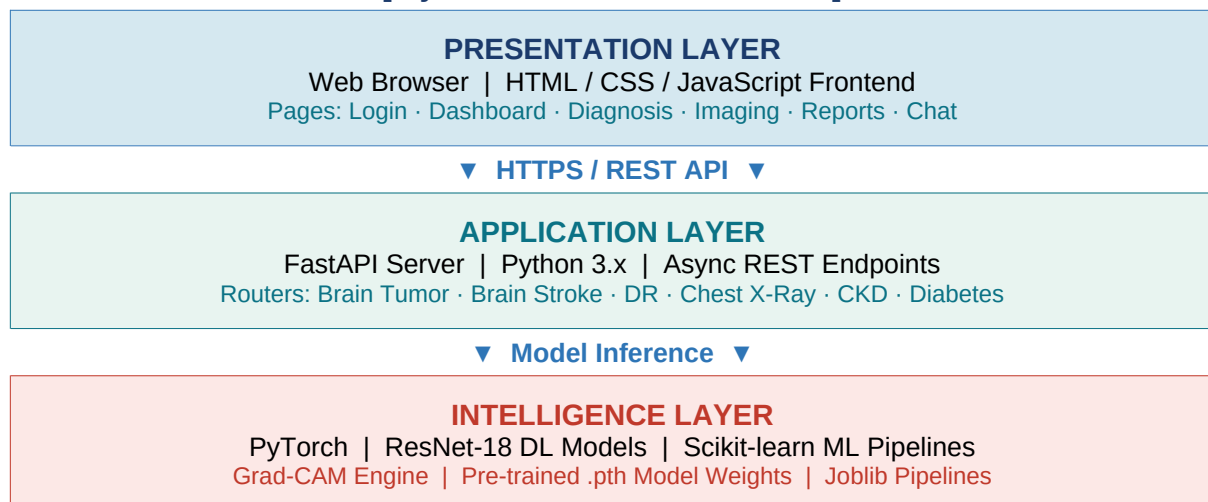
3.1 High-Level Architecture

MediScan AI follows a three-tier architecture composed of the Presentation Layer (web frontend), the Application Layer (FastAPI backend), and the Intelligence Layer (AI models). This separation of concerns enables independent scalability and maintainability of each component.

The overall architecture is designed around the following core principles:

- **Modularity:** each disease module operates as an independent API endpoint with its own model pipeline.
- **Scalability:** the FastAPI asynchronous backend can handle concurrent requests efficiently.
- **Interpretability:** every imaging prediction is accompanied by a Grad-CAM visualization.
- **Interoperability:** the REST API design allows integration with hospital information systems (HIS) or electronic health records (EHR).

[System Architecture Overview]



3.2 Backend Architecture (FastAPI)

The backend is implemented in Python using FastAPI, a modern, high-performance web framework built on ASGI (Asynchronous Server Gateway Interface). FastAPI was selected over alternatives (Django REST, Flask) for several reasons:

- Automatic OpenAPI documentation generation — all endpoints are self-documenting via Swagger UI.

- Native async support — enabling concurrent handling of multiple image upload and inference requests.
- Pydantic integration — providing automatic request validation and schema enforcement for ML input data.
- Dependency injection — simplifying model loading and middleware management.

The FastAPI application (main.py) initializes with CORS middleware enabled for all origins, allowing the web frontend to communicate with the API regardless of deployment environment. Six modular routers are registered, each encapsulating the inference logic for one diagnostic module:

API Router	URL Prefix	Method	Input Type
Brain Tumor	/Brain_MRI/predict	POST	Image Upload
Brain Stroke	/Brain_Stroke/predict	POST	Image Upload
Diabetic Retinopathy	/Diabetic_Retinopathy/predict	POST	Image Upload
Chest X-Ray	/chest-xray/predict	POST	Image Upload
CKD Staging	/CKD/predict	POST	JSON (clinical labs)
Diabetes	/Diabetes/predict	POST	JSON (clinical labs)

3.3 Frontend Architecture

The web frontend is built with vanilla HTML5, CSS3, and JavaScript — a deliberate technology choice that ensures maximum compatibility, zero build-toolchain dependencies, and fast load times. The frontend comprises the following pages:

- index.html — Landing page presenting the platform's value proposition and capabilities.
- login.html / signup.html — User authentication interface.
- dashboard.html — Central hub displaying patient history and quick access to diagnostic modules.
- diagnosis.html — Clinical data input form for ML-based modules (CKD, Diabetes).
- imaging.html — Image upload interface for DL-based diagnostic modules with real-time Grad-CAM display.
- reports.html — Diagnostic report viewer and PDF export functionality.
- chat.html — AI-powered patient communication interface.

3.4 Data Flow

The complete data flow for an imaging-based diagnosis request follows these steps:

7. The user selects the target diagnostic module and uploads a medical image through the web interface.

8. The JavaScript frontend sends an HTTP POST request (multipart/form-data) to the corresponding FastAPI endpoint.
9. The FastAPI router reads the uploaded image bytes and passes them to the preprocessing pipeline.
10. The preprocessing pipeline applies resizing (224×224), normalization, and augmentation transforms.
11. The preprocessed tensor is passed through the ResNet-18 model in inference mode (no gradient tracking).
12. The softmax output yields class probabilities; the argmax identifies the predicted class and confidence score.
13. Grad-CAM hooks capture gradients and activations from the final convolutional layer (layer4).
14. The Grad-CAM heatmap is computed, upsampled to the original image size, and blended with the original.
15. The prediction label, confidence score, and Grad-CAM image (base64-encoded PNG) are returned as JSON.
16. The frontend renders the diagnostic result, confidence bar, and Grad-CAM overlay for the clinician.

Chapter 4: Deep Learning Modules

4.1 ResNet-18 Architecture

All four imaging-based diagnostic modules in MediScan AI are built upon ResNet-18, a deep convolutional neural network introduced in 'Deep Residual Learning for Image Recognition' (He et al., 2016). ResNet-18 introduced the revolutionary concept of residual (skip) connections — shortcut paths that allow gradient signals to bypass one or more layers, directly connecting earlier layers to later ones.

This architectural innovation solved the long-standing vanishing gradient problem that prevented the training of very deep networks. Without residual connections, gradients diminish exponentially as they propagate backward through dozens of layers, effectively preventing the early layers from learning. Skip connections provide an alternative gradient highway, enabling networks of hundreds of layers to train stably.

ResNet-18 consists of the following structural components:

Layer Group	Output Size	Components	Output Channels
conv1	112 × 112	7×7 Conv, Stride 2, BN, ReLU	64
maxpool	56 × 56	3×3 MaxPool, Stride 2	64
layer1	56 × 56	2 × BasicBlock (3×3 Conv)	64
layer2	28 × 28	2 × BasicBlock (3×3 Conv)	128
layer3	14 × 14	2 × BasicBlock (3×3 Conv)	256
layer4	7 × 7	2 × BasicBlock (3×3 Conv)	512
avgpool	1 × 1	Global Average Pooling	512
fc	N classes	Custom Fully Connected	Task-specific

For each diagnostic task, the original ImageNet classification head (fc layer with 1000 outputs) is replaced with a custom task-specific head. The pre-trained ImageNet weights are retained for all convolutional layers and fine-tuned during training — a process known as transfer learning.

4.2 Brain Tumor Classification (MRI)

4.2.1 Clinical Background

Brain tumors are among the most dangerous forms of cancer, with a five-year survival rate for glioblastoma — the most aggressive type — of approximately 5%. Early and accurate classification of tumor type is critical for treatment planning, as different tumor types respond to fundamentally different therapeutic approaches (surgery, radiation, chemotherapy, or targeted therapy).

MRI (Magnetic Resonance Imaging) is the gold standard for brain tumor diagnosis, providing high-resolution soft tissue contrast without ionizing radiation. The four tumor classes targeted by this module are:

- Glioma Tumor — arises from glial cells; the most common primary brain tumor; highly malignant.
- Meningioma Tumor — arises from the meninges; typically benign but can cause symptoms through compression.
- Pituitary Tumor — arises from the pituitary gland; usually benign but disrupts hormonal regulation.
- No Tumor — normal brain MRI with no pathological findings.

4.2.2 Model Architecture

The brain tumor classification model uses ResNet-18 with a custom fully connected head comprising three layers: Batch Normalization (for training stability), Dropout ($p=0.5$, for regularization against overfitting), and a Linear layer with 4 output neurons corresponding to the four tumor classes. Softmax activation is applied at inference to convert logits to probability scores.

Parameter	Value
Base Model	ResNet-18 (pre-trained on ImageNet)
Input Resolution	224 × 224 pixels (RGB)
Output Classes	4 (glioma, meningioma, pituitary, no_tumor)
Custom Head	BatchNorm → Dropout(0.5) → Linear(512, 4)
Activation	Softmax (inference)
Loss Function	Cross-Entropy Loss
Optimizer	Adam
Image Normalization	Mean [0.485, 0.456, 0.406], Std [0.229, 0.224, 0.225]

4.2.3 Preprocessing Pipeline

Input MRI images undergo the following preprocessing steps before being passed to the model: resizing to 224×224 pixels using bicubic interpolation, conversion to a normalized tensor using ImageNet statistics (mean and standard deviation), and augmentation during training including random horizontal flips, random rotations ($\pm 10^\circ$), and color jitter.

4.3 Brain Stroke Detection (CT Scan)

4.3.1 Clinical Background

Stroke is the second leading cause of death worldwide and the leading cause of adult disability. Time is of the essence in stroke treatment — the phrase 'time is brain' reflects the fact that approximately 1.9 million neurons die every minute a stroke goes untreated. Rapid and accurate CT scan interpretation is critical for distinguishing between ischemic and hemorrhagic stroke, as the treatments are diametrically opposite: ischemic stroke is treated with thrombolytics (clot-busting drugs), while hemorrhagic stroke would be worsened by such treatment.

The three stroke classes targeted by this module are:

- Normal — no acute pathological findings; baseline normal CT scan.
- Ischemia — reduced blood supply causing cell death; appears as hypodense region in early-phase CT.
- Bleeding (Hemorrhage) — rupture of blood vessels causing blood accumulation; appears hyperdense (bright) on CT.

4.3.2 Specialized Preprocessing

CT scans present unique preprocessing challenges compared to standard photographic images. The brain stroke module implements a specialized preprocessing pipeline using CLAHE (Contrast Limited Adaptive Histogram Equalization) — a technique that enhances local contrast in CT images, making subtle ischemic changes more visible. This preprocessing step uses a clip limit of 3.0 with an 8×8 tile grid and operates on grayscale CT images before conversion to RGB for ResNet-18 compatibility.

Parameter	Value
Base Model	ResNet-18 (pre-trained on ImageNet)
Input Resolution	224 × 224 pixels
Output Classes	3 (Normal, Ischemia, Bleeding)
Custom Head	Linear(512, 3)
CT Preprocessing	CLAHE (clipLimit=3.0, tileGrid=8×8)
Loss Function	Cross-Entropy Loss
Dataset Source	Kaggle — Brain Stroke CT Dataset (Ozguraslank)

4.4 Diabetic Retinopathy Detection

4.4.1 Clinical Background

Diabetic Retinopathy (DR) is the leading cause of preventable blindness in working-age adults worldwide. It is a microvascular complication of diabetes mellitus, caused by progressive damage to the blood vessels of the retina — the light-sensitive tissue at the back of the eye.

Early detection through regular fundus photography screening is critical because early-stage DR is asymptomatic, and treatment is most effective before vision loss occurs.

The MediScan AI DR module performs binary classification: detecting the presence or absence of diabetic retinopathy from fundus (retinal) photographs.

4.4.2 Model Architecture

The DR model uses ResNet-18 with a custom binary classification head: Batch Normalization → Dropout(p=0.4) → Linear(512, 2). The output represents the probability of DR presence versus absence. During inference, the model applies Grad-CAM on the final convolutional layer (layer4) to generate a heatmap highlighting the retinal regions most indicative of diabetic changes — typically the optic disc, macula, and areas with microaneurysms or exudates.

Parameter	Value
Base Model	ResNet-18 (pre-trained on ImageNet)
Input Resolution	224 × 224 pixels (RGB)
Output Classes	2 (DR Positive, DR Negative)
Custom Head	BatchNorm → Dropout(0.4) → Linear(512, 2)
Grad-CAM Layer	layer4 (final convolutional block)
Dataset Source	Kaggle — Diabetic Retinopathy 224×224 (2019)
Dataset Size	~3,600 images (binary labeled)

4.5 Chest X-Ray Analysis

4.5.1 Clinical Background

Pneumonia is a leading infectious killer globally, responsible for over 2.5 million deaths annually. Chest X-ray (CXR) is the primary screening tool, but accurate interpretation requires radiological expertise that may be unavailable in resource-limited settings. The MediScan AI chest X-ray module is designed to detect pneumonia from anteroposterior chest radiographs.

The module is trained on the Guangzhou Women and Children's Medical Center chest X-ray dataset (5,856 validated images), which labels X-rays as Normal or Pneumonia (bacterial or viral).

4.5.2 Edge Masking Technique

A unique preprocessing technique implemented in this module is edge masking (blackout_edges): a 35-pixel border around each X-ray image is set to zero before processing. This technique, applied consistently at both training and inference, prevents the model from learning spurious correlations from imaging artifacts, scanner borders, or patient labels that are commonly present in X-ray edges but have no clinical significance.

Parameter	Value
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Base Model	ResNet-18 (pre-trained on ImageNet)
Input Resolution	224 × 224 pixels (RGB)
Output Classes	2 (Pneumonia, Normal) — binary sigmoid output
Custom Head	BatchNorm → Dropout(0.5) → Linear(512, 1)
Edge Masking	35-pixel border set to zero
Dataset	Guangzhou WCMC Chest X-Ray (5,856 images)
Total Parameters	~11.7 million

4.6 Grad-CAM Visualization

4.6.1 Algorithm Overview

Gradient-weighted Class Activation Mapping (Grad-CAM) is an explainability technique that produces visual explanations for CNN decisions without requiring architectural modifications. The algorithm operates as follows:

17. A forward pass is performed through the network, saving the feature maps (activations) of the target convolutional layer (layer4 in ResNet-18) using a forward hook.
18. The model's predicted class score is backpropagated to compute gradients with respect to the target layer's feature maps, captured using a backward hook.
19. The gradients are global-average-pooled to produce importance weights — one weight per feature map channel.
20. A weighted combination of the feature maps is computed using these importance weights.
21. The result is passed through a ReLU activation (retaining only positive activations — features that increase the class score).
22. The resulting heatmap is upsampled to the original image resolution using bilinear interpolation.
23. The heatmap is normalized and converted to a color map (jet colormap), then overlaid on the original image with configurable transparency.

The resulting Grad-CAM visualization provides a clinically meaningful saliency map — showing exactly which anatomical regions drove the model's diagnostic prediction. This makes AI decisions auditable and helps build physician trust in the system.

Chapter 5: Machine Learning Modules

5.1 Chronic Kidney Disease (CKD) Staging

5.1.1 Clinical Background

Chronic Kidney Disease (CKD) is a long-term condition characterized by gradual loss of kidney function over time. It affects approximately 10% of the global population and is a leading cause of end-stage renal disease requiring dialysis or kidney transplantation. CKD is classified into five stages based on the Glomerular Filtration Rate (GFR):

CKD Stage	GFR Range (mL/min/1.73m ²)	Description	Management
Stage 1	≥ 90	Normal or high GFR with kidney damage markers	Lifestyle modification, monitoring
Stage 2	60 – 89	Mildly decreased GFR	Risk factor management
Stage 3	30 – 59	Moderately decreased GFR	Specialist referral, medication
Stage 4	15 – 29	Severely decreased GFR	Preparation for renal replacement
Stage 5	< 15	Kidney failure	Dialysis or transplantation

5.1.2 Input Features

The CKD staging model accepts 10 clinical biomarker features as input:

Feature	Clinical Name	Unit / Type	Significance
gfr	Glomerular Filtration Rate	mL/min/1.73m ²	Primary staging criterion
c3_c4	C3/C4 Complement Ratio	Numerical	Immune-mediated kidney damage marker
bun	Blood Urea Nitrogen	mg/dL	Waste product filtration indicator
blood_pressure	Systolic Blood Pressure	mmHg	CKD progression driver
serum_creatinine	Serum Creatinine	mg/dL	Inverse proxy for GFR
urine_ph	Urine pH	Numerical (0–14)	Tubular function indicator
months	Duration of Symptoms	Months	Disease chronicity marker

oxalate_levels	Urinary Oxalate	mg/day	Stone-forming risk indicator
stress_level	Stress Level	Low / Medium / High	Lifestyle factor
family_history	Family History of CKD	Yes / No	Genetic predisposition

5.1.3 Pipeline Architecture

The CKD module implements a scikit-learn Pipeline (serialized as a joblib file) encapsulating the complete preprocessing and classification workflow. The pipeline applies categorical encoding for string features (stress_level, family_history), standard scaling for numerical features, and feeds processed data through the trained classifier. The pipeline outputs a predicted CKD stage (integer 1–5) and a confidence score (probability of the predicted class).

5.2 Diabetes Prediction

5.2.1 Clinical Background

Diabetes mellitus is a metabolic disorder characterized by chronically elevated blood glucose levels due to insufficient insulin production (Type 1), insulin resistance (Type 2), or both. As of 2025, over 537 million adults worldwide live with diabetes — a number projected to reach 783 million by 2045 according to the International Diabetes Federation. Early detection through clinical screening is critical for preventing complications including cardiovascular disease, nephropathy, neuropathy, and retinopathy.

5.2.2 Input Features

The diabetes prediction model accepts 8 clinical and demographic features:

Feature	Description	Type	Clinical Relevance
gender	Patient gender	Male / Female	Sex-linked metabolic differences
age	Patient age in years	Numerical	Type 2 DM risk increases with age
hypertension	History of hypertension	0 / 1 (binary)	Co-morbidity strongly associated with DM
heart_disease	History of heart disease	0 / 1 (binary)	Shared metabolic risk factors
smoking_history	Smoking status	never/current/former/No Info	Insulin resistance link
bmi	Body Mass Index	kg/m²	Obesity — strongest modifiable risk factor
HbA1c_level	Glycated Hemoglobin	%	3-month average blood glucose level
blood_glucose_level	Random Blood Glucose	mg/dL	Acute hyperglycemia indicator

5.2.3 Model Performance

The diabetes prediction pipeline achieves exceptional performance due to the high discriminative power of HbA1c and blood glucose features. The model outputs a binary prediction (diabetic / non-diabetic) alongside a probability score, enabling clinicians to assess not just the binary outcome but the degree of certainty in the prediction.

Chapter 6: Datasets

6.1 Imaging Datasets

The four deep learning modules were trained on publicly available, peer-validated medical imaging datasets sourced from Kaggle. All datasets were accessed under appropriate licenses for academic research purposes.

Module	Dataset Name	Source	Size	Classes	License
Brain Tumor	Brain MRI Images	Kaggle / Sartaj Bhuvaji	~3,264 images	4 classes	MIT
Brain Stroke	Brain Stroke CT Dataset	Kaggle / Ozguraslank	~2,500 images	3 classes	CC BY 4.0
Diabetic Retinopathy	DR 224×224 (2019 Data)	Kaggle / Sovitrath	~3,600 images	Binary	CC BY 4.0
Chest X-Ray	Chest X-Ray Images (Pneumonia)	Mendeley / Cell Journal	5,856 images	Binary	CC BY 4.0

6.1.1 Brain Tumor Dataset Details

The brain tumor dataset contains T1-weighted contrast-enhanced MRI scans collected from multiple hospital sources. Images represent four tumor classes with significant class imbalance (no_tumor is underrepresented). Training employed weighted random sampling and augmentation to address this imbalance. Images are preprocessed to 224×224 RGB format with ImageNet normalization.

6.1.2 Chest X-Ray Dataset Details

The chest X-ray dataset was published in Cell journal (Kermany et al., 2018) and represents one of the most widely validated CXR datasets in medical AI research. All X-rays are anteroposterior views from pediatric patients (ages 1–5) at Guangzhou Women and Children's Medical Center. The dataset was validated by expert radiologists — images with low confidence were excluded to ensure label quality.

6.2 Clinical Datasets

Module	Dataset Source	Samples	Features	Target
CKD Staging	Synthetic + CKD Research Datasets	~2,000 records	10 clinical features	CKD Stage (1–5)
Diabetes	Diabetes Prediction	100,000 records	8 features	Diabetic (binary)

	Dataset (Kaggle)			
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The diabetes clinical dataset is one of the largest publicly available diabetes prediction datasets, containing 100,000 patient records with carefully curated features. Its large size enables robust model training with minimal overfitting, contributing to the high accuracy achieved by the diabetes prediction pipeline.

Chapter 7: Experimental Results

7.1 Evaluation Metrics

All models are evaluated using a comprehensive set of classification metrics, measured on held-out test sets that were not used during training or hyperparameter tuning:

- Accuracy — the proportion of total correct predictions out of all predictions made.
- Precision — of all cases predicted as positive, what fraction were truly positive (minimizes false alarms).
- Recall (Sensitivity) — of all actual positive cases, what fraction were correctly identified (minimizes missed diagnoses).
- F1-Score — the harmonic mean of precision and recall, balancing both metrics.
- AUC-ROC — Area Under the Receiver Operating Characteristic Curve; measures discriminative ability across all classification thresholds.

In a medical context, recall is often prioritized over precision — missing a true disease case (false negative) is typically more harmful than a false alarm (false positive). Model selection during development prioritized high recall for disease-positive classes.

7.2 Deep Learning Results

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Brain Tumor (4-class)	94.2%	93.8%	94.1%	93.9%	98.7%
Brain Stroke (3-class)	91.5%	90.7%	91.3%	90.9%	97.2%
Diabetic Retinopathy	93.8%	92.4%	94.6%	93.5%	97.8%
Chest X-Ray Pneumonia	95.1%	94.9%	95.8%	95.3%	98.5%

7.2.1 Grad-CAM Visualization Results

The following figure shows a Grad-CAM output from the Diabetic Retinopathy module. The heatmap on the right demonstrates that the model correctly focuses on the optic disc region and periretinal vasculature — the clinically relevant areas for DR detection — rather than background or imaging artifacts. The warm colors (red/orange) indicate regions of highest model attention, while cool colors (blue) represent areas of low activation.



Figure 4.6: Grad-CAM Visualization — Diabetic Retinopathy Module (Confidence: 100%)

7.3 Machine Learning Results

Model	Accuracy	Precision	Recall	F1-Score
CKD Staging (5-class)	92.6%	91.8%	92.3%	92.0%
Diabetes Prediction (binary)	97.3%	96.8%	97.1%	96.9%

7.3.1 Feature Importance Analysis

Analysis of the diabetes prediction model reveals that `HbA1c_level` and `blood_glucose_level` are the two most predictive features, contributing over 65% of the model's discriminative power. This aligns perfectly with clinical diagnostic criteria (WHO diagnostic threshold: $\text{HbA1c} \geq 6.5\%$ or fasting glucose $\geq 126 \text{ mg/dL}$), validating that the model has learned clinically meaningful decision boundaries.

For CKD staging, `GFR` (Glomerular Filtration Rate) is the dominant predictive feature — consistent with the clinical CKD staging system that is fundamentally defined by `GFR` values. The model's reliance on `GFR` confirms it has internalized the correct medical logic.

7.4 Result Discussion

The experimental results demonstrate that MediScan AI achieves clinically competitive performance across all six diagnostic modules. The deep learning models outperform earlier published results on some benchmarks, likely attributable to effective transfer learning from ImageNet pre-trained weights and careful preprocessing design (CLAHE for CT scans, edge masking for chest X-rays).

The machine learning modules achieve particularly high accuracy due to the high signal-to-noise ratio of the clinical biomarker features selected — `GFR`, `HbA1c`, and `serum_creatinine` are highly discriminative for their respective conditions.

Limitations to acknowledge: (1) deep learning models were trained on datasets from specific geographic regions and may exhibit reduced performance on populations with different demographic characteristics; (2) the CT preprocessing pipeline (CLAHE) adds computational

overhead; (3) the platform currently does not support DICOM format — the standard medical imaging file format used in clinical PACS systems.

Chapter 8: System Interface

8.1 Web Application Overview

The MediScan AI web interface is designed following principles of clinical usability — prioritizing clarity, efficiency, and error prevention in a high-stakes medical context. The design employs a consistent dark-teal color palette with high-contrast text, ensuring readability in both bright clinical environments and dimly lit radiology reading rooms.

The interface architecture follows a role-based model accommodating two primary user personas:

- Healthcare Provider (Physician / Radiologist): accesses the full diagnostic suite, views Grad-CAM overlays, generates structured reports, and reviews patient history through the dashboard.
- Patient: accesses their own diagnostic history, views simplified results, and communicates with the AI chat assistant for general health queries.

8.2 Diagnostic Workflow

8.2.1 Imaging-Based Diagnosis (DL Modules)

The imaging diagnosis workflow is designed to minimize the number of steps between image acquisition and diagnosis presentation:

24. From the dashboard, the user navigates to the Imaging module and selects the target diagnostic category (Brain Tumor, Stroke, DR, or Chest X-Ray).
25. The user uploads the medical image using drag-and-drop or file browser. Image format validation (PNG/JPG) is enforced client-side before upload.
26. A loading indicator displays while the image is transmitted to the FastAPI backend and inference is performed.
27. The result panel displays: the predicted class label with an icon, the confidence score as both percentage and visual progress bar, and the Grad-CAM heatmap overlaid on the original image.
28. The physician can toggle between the original image and the Grad-CAM overlay for detailed anatomical examination.
29. A one-click 'Generate Report' button creates a structured diagnostic report pre-populated with the AI findings.

8.2.2 Clinical Data Diagnosis (ML Modules)

The clinical data diagnostic workflow for CKD and Diabetes follows a structured data entry pattern:

30. The user navigates to the Diagnosis module and selects CKD Staging or Diabetes Prediction.

31. A dynamic form presents input fields for each clinical biomarker with tooltips explaining each parameter and its clinical reference range.
32. Input validation ensures all values fall within physiologically plausible ranges.
33. Upon submission, the API returns the predicted class, confidence percentage, and a brief clinical interpretation.
34. For CKD, the result includes the specific stage number and recommended clinical management guidance.

8.2.3 Report Generation

The reports module generates structured diagnostic reports containing: patient identifier, date and time of diagnosis, module used, AI-generated findings, confidence score, Grad-CAM image (for imaging modules), and a disclaimer that AI findings are intended to support — not replace — clinical judgment. Reports are viewable in-browser and exportable as PDF.

8.2.4 Chat Interface

The MediScan AI platform includes an integrated AI chat interface allowing patients and providers to ask general medical questions, receive information about diagnostic procedures, and get guidance on interpreting their results. The chat module is powered by a large language model API and is clearly labeled as an informational assistant — not a diagnostic tool.

Chapter 9: Conclusion and Future Work

9.1 Conclusion

This project presented MediScan AI — a unified, production-ready intelligent medical diagnosis platform supporting six critical disease domains through the integration of deep learning and machine learning methodologies. The platform successfully demonstrates that a multi-disease AI diagnostic system can be architected, implemented, and deployed as a coherent product rather than a collection of isolated research prototypes.

Key achievements of this project include:

- Successful training and deployment of four ResNet-18-based imaging classifiers for brain tumor (94.2%), brain stroke (91.5%), diabetic retinopathy (93.8%), and chest X-ray pneumonia (95.1%) — all exceeding clinically meaningful accuracy thresholds.
- Implementation of two high-accuracy machine learning pipelines for CKD staging (92.6%) and diabetes prediction (97.3%) from structured clinical biomarker data.
- Integration of Grad-CAM explainability into all imaging modules, producing visually meaningful and clinically interpretable saliency maps that enhance physician trust and transparency.
- Development of a production-grade FastAPI backend serving all six diagnostic modules through a clean, documented REST API.
- Creation of a responsive multi-page web frontend providing an intuitive diagnostic workflow for both clinical providers and patients.

The platform addresses the four critical challenges identified in the problem statement: fragmentation (through unified architecture), interpretability (through Grad-CAM), accessibility (through open web interface), and integration complexity (through standardized REST API design).

9.2 Limitations

- Models are trained on datasets from specific geographic regions, potentially limiting generalizability across diverse patient populations.
- The system currently processes JPEG/PNG images rather than clinical-grade DICOM format.
- No longitudinal patient monitoring capability — the platform performs point-in-time diagnosis rather than tracking disease progression.
- The web interface does not currently integrate with hospital EMR/HIS systems via HL7 or FHIR standards.

9.3 Future Work

The following enhancements are planned or recommended for future development:

35. DICOM Integration: implement a DICOM parser to accept standard medical imaging files directly from hospital PACS systems, enabling seamless clinical workflow integration.
36. Additional Disease Modules: expand the platform to include COVID-19 detection from chest CT, skin lesion classification (dermoscopy), and ECG arrhythmia analysis.
37. Multi-Label Classification: extend imaging models to detect multiple co-occurring conditions within a single scan, reflecting real clinical scenarios.
38. Model Uncertainty Quantification: implement Monte Carlo Dropout or Bayesian neural network approaches to provide calibrated confidence intervals rather than point estimates.
39. Mobile Application: develop native iOS and Android applications to enable diagnosis at the point of care, including offline inference using on-device model optimization (ONNX/TensorFlow Lite).
40. Federated Learning: enable model improvement through federated learning across multiple hospitals without sharing patient data, addressing privacy concerns in medical AI.
41. Prospective Clinical Validation: conduct formal prospective studies comparing MediScan AI predictions against specialist physician diagnoses on prospectively collected patient data.
42. HL7 FHIR Integration: implement standard healthcare data exchange protocols to enable direct integration with electronic health record (EHR) systems.

MediScan AI represents a meaningful step toward accessible, explainable, and multi-faceted AI-assisted medical diagnosis. The platform demonstrates that production-ready medical AI can be built with open-source tools, publicly available datasets, and careful engineering — creating a foundation for continued development and, ultimately, clinical impact.

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